

CoLoca-QuA: Cooperative Relative Pose Estimation With Query-Based Attention Using Neural Intermediate Features in V2X Network

Letian Gao ^{lb}, *Member, IEEE*, Hao Xiang ^{lb}, Tianhao Tan, Zonglin Meng ^{lb}, Xin Xia ^{lb}, *Senior Member, IEEE*,
Lu Xiong ^{lb}, and Jiaqi Ma ^{lb}, *Senior Member, IEEE*

Abstract—Intelligent vehicles and transportation systems have advanced significantly in recent years, with cooperative driving automation showing great potential to further enhance vehicle intelligence and transportation safety. Accurate pose is crucial for enabling cooperative driving automation, as information from different transportation participants must be integrated into a common reference frame. However, challenges remain in achieving precise pose information. The self-localization systems of ego vehicles can fail in complex driving environments, leading to large initial pose errors in cooperative driving systems, while transmission bandwidth limitations constrain the amount of shared information. To address these issues, we propose a learning framework based on query-based attention for vehicle cooperative pose estimation using neural intermediate features (CoLoca-QuA) in Vehicle-to-Everything (V2X) network. This framework aims to estimate the relative pose error between the ego vehicle and Connected Automated Vehicles (CAVs) or intelligent infrastructures. We validate our algorithm on both the simulated *V2Xset* and the real-world *V2X-Real* datasets, and it outperforms state-of-the-art relative localization methods, demonstrating superior performance over state-of-the-art relative localization methods. Our approach not only requires a small bandwidth but also accurately estimates relative pose even in scenarios with significant initial errors, thereby enhancing downstream cooperative perception tasks.

Index Terms—Cooperative localization, connected automated vehicles, relative localization, machine learning, intermediate feature.

I. INTRODUCTION

VEHICLE technology is rapidly trending toward electrification and intelligence, and the advanced perception and execution capabilities of intelligent electric vehicles enhance driving

Received 14 August 2025; revised 21 November 2025; accepted 14 January 2026. Date of publication 20 January 2026; date of current version 20 July 2026. This work was supported in part by the National Science Funds for Distinguished Young Scholar of China under Grant 52325212, in part by the Fundamental Research Funds for the Central Universities of China, and in part by the Federal Highway Administration Center of Excellence on New Mobility and Automated Vehicle Program of the US. The review of this article was coordinated by Dr. Naoki Ishikawa. (*Corresponding author: Jiaqi Ma.*)

Letian Gao, Tianhao Tan, and Lu Xiong are with the School of Automotive Studies, Tongji University, Shanghai 201804, China.

Hao Xiang, Zonglin Meng, and Jiaqi Ma are with the UCLA Mobility Lab, Department of Civil and Environmental Engineering, UCLA, Los Angeles, CA 90095 USA (e-mail: jiaqima@ucla.edu).

Xin Xia is with the Department of Mechanical Engineering, University of Michigan, Dearborn, MI 48128 USA.

Digital Object Identifier 10.1109/TVT.2026.3656231

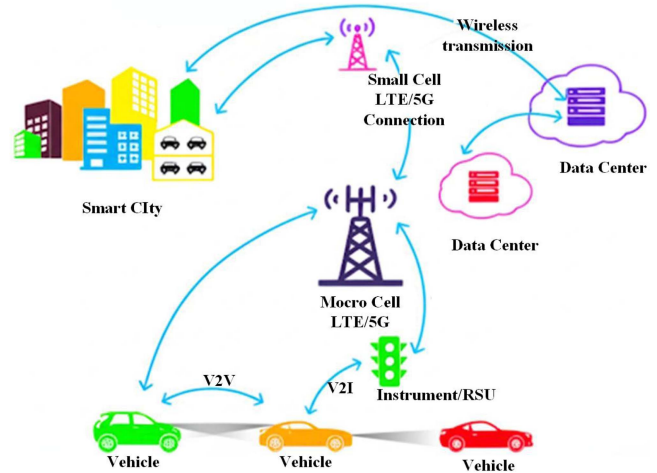


Fig. 1. V2X - Vehicle to Everything.

safety greatly. Cooperative Driving Automation (CDA) can extend the sensing range of individual traffic participants through information sharing in intelligent transportation systems [1], [2], as illustrated in Fig. 1. This enables each participant to enhance their perception capabilities, which improves overall transportation safety, efficiency [3], and resilience [4], [5]. By utilizing comprehensive data and sensor signals from multiple traffic participants, CDA enhances the accuracy and robustness of intelligent driving systems, facilitating advanced applications such as vehicle platoon control [6]. By utilizing comprehensive data and sensor signals from multiple traffic participants, CDA enhances the accuracy and robustness of intelligent driving systems.

To achieve CDA, information must be transmitted among connected traffic participants (referred to as agents) and subsequently fused at a central unit, such as a Connected Automated Vehicle (CAV) or a central processing hub. The pose information of each agent is crucial, as it enables the spatial alignment of data from different agents. In cooperative perception tasks, for instance, pose errors can diminish detection capabilities and lead to false detections. As illustrated in Fig. 2, relative pose errors may cause point cloud misalignments, where the displaced point cloud (yellow points) deviates from the ground truth point cloud (blue points) after transformation into the ego vehicle’s frame (with orange points representing the ego). This

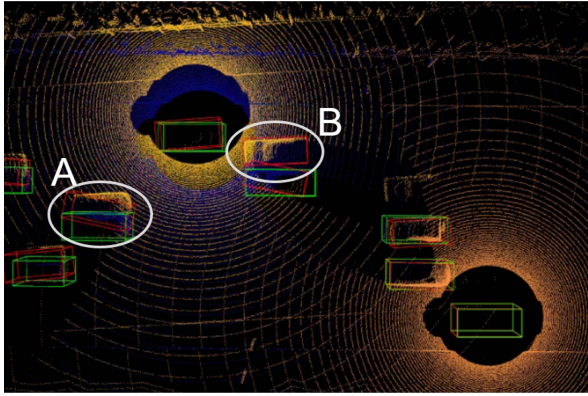


Fig. 2. The influence of localization error on detection. (Point clouds: orange from the ego vehicle, blue from the CAV and transformed using the ground truth pose, yellow from the CAV and transformed using the noisy pose. Bounding boxes: ground truth in green, detection in red.).

misalignment disrupts cooperative perception ('A' in Fig. 2, resulting in reduced detection accuracy) and can lead to false detections ('B' in Fig. 2).

Most studies assume that vehicles can obtain accurate absolute localization results from their self-localization systems, allowing for the derivation of relative pose and alignment of information from different agents. However, achieving continuous and precise localization is a significant challenge for self-localization systems in real-world driving conditions. Global Navigation Satellite Systems (GNSS) is a widely used localization system for vehicles, but its signal quality is easily affected or obstructed by environmental factors, and its update frequency is insufficient to meet the demands of real-time driving. To address these limitations, Inertial Measurement Unit (IMU) and vehicle chassis sensors are typically integrated with GNSS to enhance the continuity, accuracy, and update frequency of localization systems. Additionally, with the advancement of sensors in intelligent vehicles, techniques such as LiDAR-(Light Detection and Ranging) and camera-based map matching, Simultaneous Localization And Mapping (SLAM), and infrastructure-based methods have also been developed and incorporated into localization systems. However, these methods often rely on external data, such as pre-built maps or infrastructure, which may not be consistently available throughout the entire journey of an intelligent vehicle. As mentioned earlier, accurate relative pose is crucial for CDA. The performance of perception systems is significantly enhanced through cooperation with other connected agents, as cooperative perception effectively addresses challenges such as occlusion and limited sensing range. A more advanced perception system can provide precise and robust detection results, as well as additional data for downstream systems such as decision-making [7], path planning, and motion control, ultimately improving the safety of intelligent driving. Various fusion strategies are commonly employed to enable cooperative perception. Early fusion techniques combine raw sensor data from different agents, preserving most of the original information but requiring substantial bandwidth and potentially leading to delays and increased computational load [8]. Late fusion, on the other hand, aggregates the final processed outputs, such as bounding boxes, from other agents, which follow a

standardized transmission format and require minimal bandwidth. However, the fusion accuracy is constrained by the performance of the other agents' perception systems. A balanced approach, intermediate fusion, merges intermediate feature maps extracted by a neural network backbone from the raw sensor data of each collaborating agent [9]. This method retains much of the original data's information while reducing bandwidth requirements. Regardless of the fusion method used, accurate relative pose information between all collaborative agents is essential. Consequently, improving localization accuracy by leveraging information from other connected traffic participants through CDA has become a valuable research direction.

Given the advantages of intermediate fusion for cooperative perception, leveraging intermediate features from different agents for cooperative localization is a promising approach for establishing a unified framework for CAVs. For CAVs with intermediate fusion-based perception systems, utilizing the existing intermediate features to estimate pose information can reduce both system complexity and computational load. On the other hand, most current studies on cooperative perception operate under ideal conditions, assuming no localization errors. In such scenarios, the information from various agents aligns perfectly, resulting in accurate perception outcomes. However, in real-world applications, localization errors are inevitable and can significantly affect cooperative perception performance. While some studies have considered localization errors in cooperative perception systems [10], [11], the introduced noise is often small, typically less than half a meter. In practice, larger localization errors pose a substantial challenge. In urban environments, standard GNSS localization errors typically range from 1 to 3 meters in the absence of Real-time Kinematic (RTK) correction, primarily driven by multipath effects or signal occlusion from skyscrapers and bridges. Furthermore, if GNSS signals are fully blocked, the localization error will diverge over time due to the accumulated drift of inertial sensors. To address this, we aim to investigate whether cooperative pose estimation can be achieved through a learning-based network that leverages intermediate features and remains robust to large initial localization errors.

Moreover, to the best of our knowledge, no existing work has specifically addressed cooperative relative pose estimation using query-based attention based on intermediate feature fusion. We introduce a novel approach in this paper by utilizing query-based attention to extract more valuable global features and query the relative pose error through a Transformer-based learning network. To validate the proposed method, we not only compare its performance with CNN (Convolutional Neural Networks)-based approach [12] but also evaluate it against traditional point cloud matching methods.

The contributions of this paper can be summarized as follows:

- We propose a cooperative relative pose estimation framework based on intermediate neural feature fusion.
- With the novel approach of querying localization errors, we developed CoLoca-QuA, a learning framework that utilizes query-based attention for vehicle cooperative localization using neural intermediate features. This framework effectively addresses the challenge of large initial relative pose errors in cooperative driving environments, benefiting downstream cooperative driving tasks.

- The proposed method is validated on open-source datasets, not only on a simulated dataset, but also on a real-world dataset. Compared to the state-of-the-art methods, our approach demonstrates high accuracy while also maintaining a low bandwidth requirement.

II. RELATED WORK

A. Vehicle Localization Systems

Accurate localization is very important for autonomous driving functions. As for cooperative driving automation in the V2X network, relative pose among the agents is necessary for sensor information alignment [13]. While a rich body of work focuses on cooperative control, lane-free traffic management [14], and platoon coordination in intelligent transportation systems [15], these studies typically assume that reliable absolute or relative localization is already available as a system input. Usually there are two methods for obtaining the relative pose information, calculate by absolute localization systems from different agents, the other is to measure the relative pose among the agents directly using additional range sensors.

Obtain relative pose estimation based on absolute localization information rely on the performance of each agent's localization system [16].

The most commonly used absolute localization system on vehicles is the GNSS/INS (Inertial Navigation System) fusion system [17], which takes advantage of the high-frequency, reliable and continuous output of IMU, and also the high accuracy velocity and pose measurements from the GNSS. However, GNSS can only achieve good performance in open-sky scenarios, and in urban driving scenarios, the accumulative error of INS will cause localization results divergency during GNSS outages [18].

Although wheel speed sensor [19], LiDAR and camera based map-matching techniques [20] could compensate the error to a certain degree, it is difficult to maintain high-accuracy pose estimation results for vehicle's own absolute localization system, and in challenging driving scenarios only meter-level accuracy could be reached [21], which is far out of the requirement for CDA tasks.

Relative pose information can also be measured or estimated by using additional range sensors. UWB (Ultra Wide-Band) technique can be used for high-accuracy relative pose estimation, however, it needs UWB infrastructures and also additional labels mounted on each vehicle with the same communication protocol. LiDAR, camera, and radar [22] can also be used for relative pose estimation by using object detection techniques. Usually, the parameters of the bounding boxes of the objects show the relative position and heading angle between the objects and the ego vehicle, but the accuracy heavily relies on the performance of the detection algorithm, and the compression of the original sensor information may decrease the accuracy on the other hand.

B. LiDAR-Based Perception Systems

LiDAR could provide point cloud of the environment with accurate 3-D positions, which impressively improves the

perception performance of the vehicle [23]. Based on LiDAR raw point cloud measurement, feature extraction networks are developed to conduct detection and other environment perception tasks. PointNet is used for feature extraction directly from 3-D point cloud [24]. Voxelnet [25] divides point cloud into voxels, and then extract features based on voxels, therefore heavily decrease the computation load of 3-D point cloud processing. SECOND [26] further improves the efficiency and accuracy of Voxelnet by modifying the loss function. But 3-D convolutional layers are necessary for the above mentioned methods, which demands more computation resources. To reduce the computation load, Bird's Eye View (BEV) representations are proposed [27]. Using BEV, PointPillar [28] is proposed to transform 3-D point cloud into 2-D representation, and then PointNet is used for feature extraction. After feature extraction, a regression head and a classification head are usually used for object detection and classification.

C. Cooperative Perception Considering Relative Pose Error

The capacity of ego vehicle perception system has limitations in scenarios that occlusion occurs. In such situations, cooperative perception shows great potential in expanding the perception range and improving the robustness and accuracy of object detection by fusing information from multiple agents in the V2X network. To fuse the sensor information from different agents, the information needs to be transformed into the same frame, therefore, relative pose among the agents is very important for measurements alignment, and the localization error directly influences the performance of cooperative perception system. According to the analysis in [29], a half meter position error and 0.1 deg heading angle error will result in over 30% accuracy reduction in cooperative detection task. In existing cooperative perception studies, most compensate the localization error before the fusion process, or simply assume that the localization error is sufficiently small and can be ignored.

Despite the relative pose estimation methods based on ego vehicle, information exchange among agents in the same communication network can also be used for enhancing the relative pose estimation accuracy [30]. Without adding additional equipments such as UWB on the intelligent vehicles, methods sharing more common sensor information are also studied recently. The agents in the V2X network could share the detected bounding boxes of the objects, then the bounding boxes from different agents could be aligned using coarse initial relative poses among agents, which can be obtained through ego-localization system with pose errors. Using bounding boxes alignment techniques [10], the initial relative pose error could be estimated and then corrected, however, this method require that the initial error is not large, and the accuracy is also limited by the performance of the detection algorithms. In addition, bounding-box level information is a heavily compressed representation of the original measurements, which discards fine-grained geometric cues that would be beneficial for high-precision localization and makes the optimization problem more sensitive to outliers and missed detections in crowded urban scenes.

For example, the bounding boxes from different agents' detection system are aligned first to estimate the relative pose error, and then corrected relative pose is used for further cooperative information fusion [10], [11]. Other than sharing the processed bounding boxes, sharing sensor original data can eliminate the processing errors caused by the ego's algorithm, but also require more transmission bandwidth and computation resources. Dong et al. estimate the relative pose among vehicles by sharing the LiDAR original raw points [31]. To reduce the transmission and computation burden, Gao et al. proposed to extract feature points from raw points, and only feature points are transmitted and used for relative pose estimation [32]. These point-cloud-matching approaches can, in principle, converge to highly accurate relative poses given a moderate initial error and a sufficiently large overlapping area. However, they typically rely on iterative optimization, which introduces significant computational overhead and may diverge in the presence of large initial misalignment.

Without correcting the relative pose in advance, a pose error rectification module based on feature maps matching technique is used to match the feature map processed by the backbone and correct the pose error before information fusion in [33]. [34] integrate a relative pose regression module directly into an end-to-end cooperative perception network. A key advantage of using intermediate feature maps as the information carrier is that the feature space is already semantically enriched, providing inherent robustness to sensor noise and partial observations. Furthermore, jointly learning pose refinement with perception tasks allows the system to directly optimize for final application performance. However, a common limitation is that these methods treat pose estimation as an auxiliary component within a specific perception network. Consequently, localization performance is typically evaluated only indirectly through downstream detection metrics, rather than being systematically analyzed as a standalone cooperative localization problem.

Moreover, existing learning-based pose refinement methods are typically validated under relatively mild localization errors, which corresponds to already well-calibrated ego localization. These observations motivate a more general cooperative localization framework that: (i) operates directly on shared intermediate neural features to respect bandwidth constraints and reuse existing perception backbones; (ii) explicitly targets the estimation of relative pose residuals as a main objective, rather than a byproduct of detection; and (iii) remains robust under large initial pose errors which better reflect real-world V2X deployment scenarios. From this perspective, our CoLoca-QuA framework can be viewed as extending the line of intermediate-feature-based cooperative perception to the domain of cooperative localization, and filling the gap between high-bandwidth point-cloud matching methods and detection-oriented feature rectification modules that only handle small localization disturbances.

III. METHODOLOGY

A. Overall Structure

To execute cooperative driving tasks, information from different agents needs to be transformed into a common reference

frame using localization data. This paper aims to achieve accurate relative pose estimates to effectively align information from multiple agents. The overall structure of the proposed localization method is shown in Fig. 3.

Assuming the vehicle operates on a planar surface, which is a condition consistent with most urban driving environments, then the vertical displacement can be neglected. Furthermore, since the vehicle's pitch and roll angles are typically within 5° , their contribution to the frame transformation matrix is negligible. Consequently, the vehicle pose can be simplified to a three-degree-of-freedom representation: translation along the X and Y axes, and rotation around the Z-axis.

In the first step, the CAVs receive the ego vehicle's pose, compute the relative pose, and transform their raw LiDAR data into the ego frame. We assume each agent is provided with an initial (noisy) absolute pose obtained from onboard localization systems (e.g., GNSS/INS fusion or SLAM) within the world frame. Denoting this world frame as W , let the estimated absolute poses of agent i and agent j be T_i^W and T_j^W , respectively.

We represent an agent pose as a homogeneous transform; unless otherwise stated, we use the Special Euclidean group $SE(3)$, which describes rigid body transformations (rotation and translation) in 3D space:

$$T = \begin{bmatrix} R & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}, \quad R \in SO(3), \mathbf{t} \in \mathbb{R}^3,$$

where $SO(3)$ (Special Orthogonal group) represents the rotation component. The relative transform from agent j to ego agent i is then

$$T_j^i = (T_i^W)^{-1} T_j^W. \quad (1)$$

Because downstream alignment and fusion are carried out in BEV, we frequently work with the planar specialization in $SE(2)$, which simplifies the motion to a 2D plane (translation in x, y and rotation in yaw ψ):

$$T_{SE(2)} = \begin{bmatrix} \cos \psi & -\sin \psi & x \\ \sin \psi & \cos \psi & y \\ 0 & 0 & 1 \end{bmatrix},$$

where (x, y, ψ) denotes the pose on the ground plane. When point-cloud operations require it, the $SE(2)$ update is embedded back into $SE(3)$.

This T_j^i inherently carries the combined localization errors from both agents' initial estimates. It is used to project agent j 's LiDAR points into the ego frame:

$$D_j^i = T_j^i D_j. \quad (2)$$

Afterward, the transformed point cloud is cropped to the ego's sensing range to keep only the overlapping area.

In the second step, intermediate feature representations are obtained from different agents. Both the CAV and the ego apply PointPillars [28] to extract BEV feature maps. Denoting the backbone by $f(\cdot)$,

$$F_j^i = f(D_j^i), \quad F_i^i = f(D_i^i), \quad (3)$$

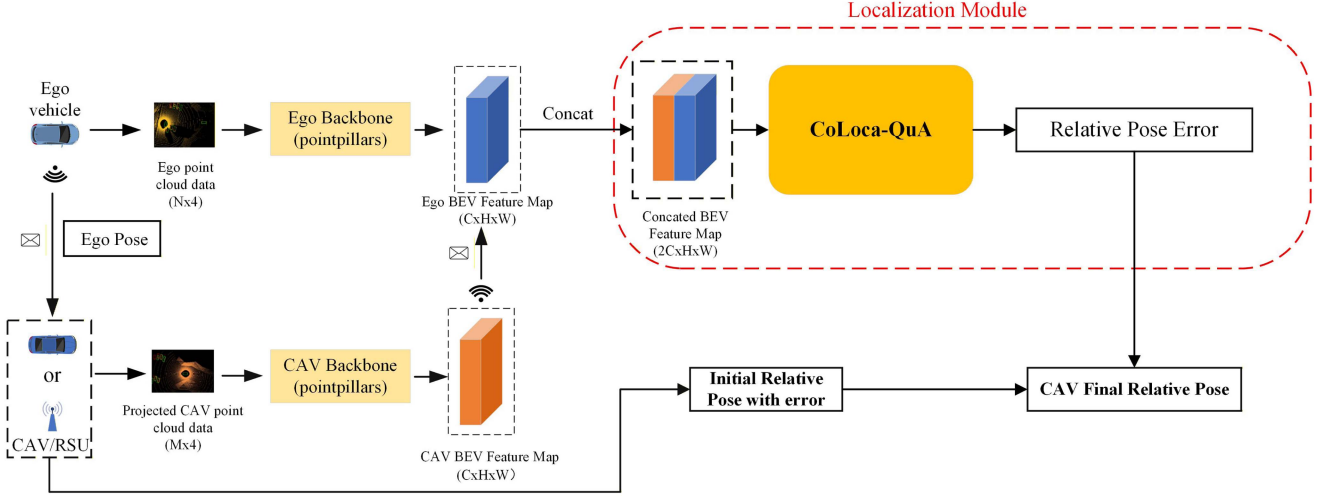


Fig. 3. Overall structure of the cooperative localization method.

where $F \in \mathbb{R}^{C \times H \times W}$ is the BEV feature with channels C , height H , and width W . The non-ego feature F_j^i is transmitted to the ego and concatenated with F_i^i .

In the third step, we employ a Transformer-based localization module (CoLoca-QuA) to estimate the residual relative pose error δ from the concatenated features:

$$\delta = l(F_j^i, F_i^i), \quad (4)$$

where $l(\cdot)$ denotes the localization head. The output is a 3D error vector

$$\delta \mathbf{p}_j^i = [\Delta x, \Delta y, \Delta \psi]^T,$$

with $\Delta x, \Delta y$ the planar translations along the ego longitudinal/lateral axes and $\Delta \psi$ the yaw correction. Notation: the superscript T here denotes vector/matrix transpose, whereas the superscripts/subscripts in T_j^i indicate frames (“from j to i ”).

These parameters form a SE(2) correction

$$C_j^i = \begin{bmatrix} \cos(\Delta \psi) & -\sin(\Delta \psi) & \Delta x \\ \sin(\Delta \psi) & \cos(\Delta \psi) & \Delta y \\ 0 & 0 & 1 \end{bmatrix}. \quad (5)$$

For point-cloud operations, we embed C_j^i into SE(3) as

$$C_{j,4 \times 4}^i = \begin{bmatrix} \cos(\Delta \psi) & -\sin(\Delta \psi) & 0 & \Delta x \\ \sin(\Delta \psi) & \cos(\Delta \psi) & 0 & \Delta y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (6)$$

and obtain the refined relative pose by pre-multiplying the initial estimate:

$$T_{j,c}^i = C_{j,4 \times 4}^i T_j^i. \quad (7)$$

The corrected transform yields improved alignment when re-projecting the CAV point cloud:

$$D_{j,c}^i = T_{j,c}^i D_j. \quad (8)$$

Notably, if the absolute pose of another agent is known, the ego vehicle can, in turn, refine its own absolute pose estimate using the corrected relative transformation. This refined global localization can consequently enhance downstream autonomous driving tasks.

B. Query-Based Attention in Localization Module

The target of relative pose regression is to infer the SE(2) offset that best aligns two BEV feature maps. This is a global matching problem that benefits from modeling long-range dependencies and content-adaptive evidence aggregation. Multi-head self-attention provides exactly these capabilities: each query attends to all spatial locations and dynamically reweights them based on feature similarity, enabling robust recovery from large initial offsets. By comparison, CNN-based heads typically summarize spatial evidence using global average pooling (GAP) or global max pooling (GMP). GAP erases spatial selectivity by equally weighting all locations; GMP preserves only the peak response, making the descriptor brittle to spurious peaks and insensitive to distributed, multi-object evidence. Both are fixed, content-agnostic reducers that weaken the required “search-then-focus” behavior for alignment, especially under large initial offsets where long-range correspondences must be discovered. In our design, a learnable Localization Token serves as a task-specific query that interacts with all patches across layers to selectively collect pose-relevant cues, and its final representation is directly mapped to $[\Delta x, \Delta y, \Delta \psi]$. Therefore, after the feature extraction from the original point cloud, attention to global features is essential. Inspired by this, we propose an approach that directly queries the localization error from learnable features using a Transformer in the localization module. Our method leverages the attention mechanism to capture spatial interactions among multi-source LiDAR features, processing them into global representations for final localization refinement. The detail of the proposed localization module is shown in Fig. 4.

1) *Model Configuration*: Our localization head is a ViT-style encoder operating on BEV patches. Table I summarizes the

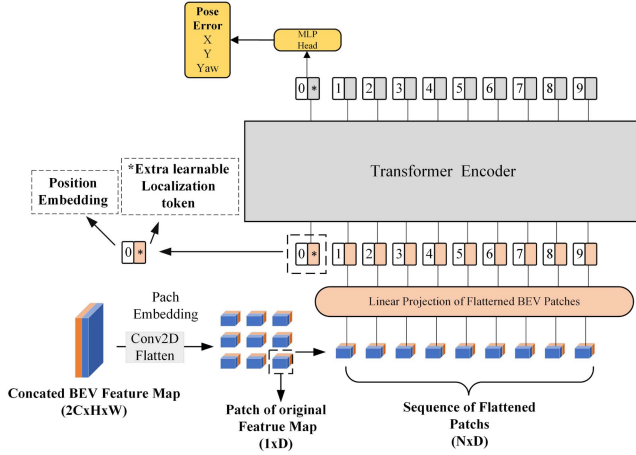


Fig. 4. Localization Module. C represents the number of channels in the feature map. H represents the height of the feature map. W represents the width of the feature map. N is the total number of patches the feature map is divided into. D is the desired dimension for each patch embedding. In our inference settings: $C=256, H=96, W=256, N=96, D=256$.

TABLE I
LOCALIZATION MODULE HYPERPARAMETER CONFIGURATION

Patch size P (pixels in BEV)	16
Embedding dim D	256
Heads	4
MLP ratio r	4
Encoder layers L	6
Dropout (attn)	0.3
Positional encoding	learnable
Loc-Token	learnable, dim D
Patch embedding	Conv- $P \times P$, stride P
Normalization	Pre-LN (LayerNorm)

configuration: patch size P , embedding dimension D , number of attention heads H , number of encoder layers L , dropout, positional encoding, and tokenization scheme. We choose a moderate patch size P to balance spatial fidelity (for large-offset alignment) and token count $N = \frac{HW}{P^2}$ for tractable attention. The embedding dimension D and head count H are set to keep the per-layer FLOPs within real-time budgets on our target GPU while preserving global matching capacity.

The original LiDAR point cloud is first voxelized, and by compressing along the z-dimension, a bird's-eye view (BEV) representation is obtained, which is then fed into the Point-Pillar backbone. This process produces a spatial feature map $F \in \mathbb{R}^{C \times H \times W}$, where C represents the channel, W the width, and H the height. Both the ego vehicle and the CAV extract their respective feature maps, denoted as F_{ego} and F_{cav} . After receiving the feature map from the CAV, the ego vehicle concatenates them along the channel dimension to form a single input for the localization module:

$$F_{\text{concat}} = \text{Concat}(F_{ego}, F_{cav}) \in \mathbb{R}^{2C \times H \times W} \quad (9)$$

Building on the idea that the relative pose error can be queried from a trained model, we design the localization module using a Transformer. The feature maps from the two vehicles are treated as sets of questions and answers; if a new query matches a

question from the training phase, the corresponding answer (relative pose error) is returned. As the encoded multi-agent LiDAR features contain different location errors, we apply self-attention to learn dependencies among these noisy feature representations and to establish the spatial relationships between agents.

First, to transform the concatenated BEV feature maps into a sequence of embeddings suitable for a Transformer, we employ a convolutional patch embedding strategy. A single 2D convolution with a kernel size and stride of P is applied to F_{concat} . The number of output filters is set to the desired embedding dimension, D . This operation efficiently maps the feature map into a grid of patches:

$$F_{\text{patched}} = \text{Conv2D}(F_{\text{concat}}) \in \mathbb{R}^{D \times \frac{H}{P} \times \frac{W}{P}} \quad (10)$$

The resulting tensor is then flattened and permuted to form a sequence of N patch embeddings, where $N = \frac{H \times W}{P^2}$:

$$Z_{\text{patches}} = \text{Permute}(\text{Flatten}(F_{\text{patched}})) \in \mathbb{R}^{N \times D} \quad (11)$$

Next, we introduce a learnable **Localization Token** ($z_{\text{loc}} \in \mathbb{R}^D$), prepending it to the patch sequence. This token is designed to aggregate global information from all patches:

$$Z_{\text{seq}} = \text{Concat}([z_{\text{loc}}], Z_{\text{patches}}, \text{axis} = 0) \in \mathbb{R}^{(N+1) \times D} \quad (12)$$

Finally, learnable **Positional Embeddings** ($E_{\text{pos}} \in \mathbb{R}^{(N+1) \times D}$) are added to the sequence to retain spatial information. The resulting sequence, $Z'_0 = Z_{\text{seq}} + E_{\text{pos}}$, serves as the input to the Transformer encoder.

After the input sequence is processed by the multi-head attention layers of the Transformer encoder, we extract the final state of the localization token, $z_{\text{loc_final}}$. This token's state, which now contains globally-aware information about the spatial misalignment, is fed into a simple MLP (linear connection layers) to output the estimated pose error vector. This vector includes the position errors in the longitudinal and lateral directions ($\Delta x, \Delta y$), as well as the yaw error ($\Delta \psi$):

$$[\Delta x, \Delta y, \Delta \psi]^T = \text{MLP}((z_{\text{loc_final}})) \quad (13)$$

IV. EXPERIMENTS

A. Dataset

The proposed method is tested on both a simulated dataset and a real-world dataset.

a) V2Xset: The V2Xset dataset [29] is used to train and validate the proposed algorithm. V2Xset is an open-source, large-scale dataset created via CARLA/OpenCDA simulation, containing two connected automated vehicles (CAVs) and one infrastructure agent. It supports cooperative perception tasks in which multiple agents share information to improve scene understanding. V2Xset spans 11,447 frames with multi-modal data per frame, including ego poses, LiDAR point clouds, camera images, and ground-truth 3D bounding boxes. In this work, we reformulate the data for cooperative localization by constructing two-agent samples for V2V and V2I settings, which are used to train and test relative-pose estimation.

b) V2X-Real: To assess real-world robustness, we further evaluate the proposed method on the V2X-Real dataset [35].

V2X-Real contains time-synchronized recordings from connected vehicles and roadside units (RSUs) in real traffic, with multi-sensor measurements (e.g., LiDAR and cameras). It provides 33,000 LiDAR frames and 171,000 multi-view camera data with ground-truth pose information. Following the same protocol as on V2Xset, we reorganize V2X-Real into V2V and V2I subsets for relative-pose estimation.

B. Implementation and Training Details

Following the default configuration of OpenCood, the evaluation LiDAR range spans $[-102.4, 102.4]$ m along the x -axis and $[-38.4, 38.4]$ m along the y -axis. The maximum communication distance between the ego vehicle and other agents is limited to 40 m. Regarding the model architecture, the proposed CoLoca-QuA is configured with 6 transformer encoder layers. For optimization, the learning rate is initialized at 1×10^{-4} .

To strike a balance between communication bandwidth and accuracy, we employ a pillar-based voxelization strategy that compresses the input point cloud along the z -dimension. Specifically, for the V2XSet dataset (32-line LiDAR), the voxel size is set to 0.2×0.2 m. In contrast, for the V2X-Real dataset equipped with higher-density 128-line LiDARs, we adopt a larger voxel size of 0.4×0.4 m to accommodate the increased point density.

During the training process, we first train the feature extraction backbone using a cooperative perception head. Next, we remove the perception head, and add the proposed CoLoca-QuA model, then fix the backbone and train the CoLoca-QuA model. After this initial training, we unfix the backbone and proceed with fine-tuning.

Specifically, the PointPillar backbone is trained using the cooperative perception model F-cooper [36]. The perception head consists of a classification head and a bounding box regression head. During this stage, the training data are free of pose errors, meaning we use the ground truth poses of the ego vehicle and the CAVs/infrastructure to perform coordinate transformations. This approach allows us to train the backbone specifically for feature extraction from a single LiDAR point cloud. Once the perception results converge, we fix the parameters of the backbone.

In the localization model training phase, we utilize the fixed backbone along with the proposed CoLoca-QuA model, while removing the perception head used during backbone training. To simulate localization errors in real-world scenarios, pose noises are manually added to the ground truth poses of CAV. As a result, the point clouds from the ego vehicle and the CAV will not align perfectly due to the existing relative pose error; however, we also possess the ground truth relative pose, allowing us to compute the localization loss. Since the backbone compresses the original 3D LiDAR point cloud into a BEV representation, our focus is solely on the 2D pose, which includes position in the longitudinal and lateral directions, as well as the heading angle. Therefore the training loss is formed as:

$$\begin{aligned} L_{loc} = & \alpha_x l_n(\delta x_{gt}, \delta x_m) \\ & + \alpha_y l_n(\delta y_{gt}, \delta y_m) \\ & + \alpha_\varphi l_n(\delta \varphi_{gt}, \delta \varphi_m) \end{aligned} \quad (14)$$

where L_{loc} is the localization loss, l_n is the MSE loss, $\delta x_{gt}, \delta y_{gt}$ and $\delta \varphi_{gt}$ are the ground truth of the relative position error in x and y directions and the heading error, $\delta x_m, \delta y_m$ and $\delta \varphi_m$ are the relative position error in x and y directions and the heading error from the trained network. $\alpha_x, \alpha_y, \alpha_\varphi$ are the weighting parameters.

Since pose noises are manually added to the ground truth poses of CAV, we applied a relatively large error to the data to assess the effectiveness of the proposed method in scenarios with significant localization errors. The added noise is randomly sampled from an error dataset that includes Gaussian noise with a mean value of $\mu = 0$ and standard deviation of $\sigma = 2$ m and $\sigma = 1$ m. Additionally, heading angle noise, with a mean value of $\mu = 0$ and a standard deviation of $\sigma = 1^\circ$, is also included. This noise is applied to the ground truth pose of the CAV (for V2V) or the ego vehicle (for V2I). In this phase, we first train the CoLoca-QuA model with the fixed backbone, followed by unfixing the backbone and fine-tuning the entire model. We utilize the Adam optimizer, and the model is trained on an NVIDIA RTX 4090D using PyTorch.

C. Evaluation Setup

Inference Settings and Evaluation Metrics: In the inference phase, we evaluated the trained model under two noise settings: Gaussian noise with a mean value of $\mu = 0$, standard deviation $\sigma = 2$ m and standard deviation $\sigma = 1$ m, respectively. Additionally, heading angle noise was introduced as Gaussian noise with a mean value of $\mu = 0$ and a standard deviation of $\sigma = 1^\circ$. These noises were added to the ground truth poses of the agents. The final localization accuracy is assessed using the mean absolute error (MAE) and the root mean squared error (RMSE), providing an overall evaluation of the proposed model's performance across the entire dataset. To further evaluate the accuracy of the proposed method at different levels, we introduce an indicator representing the percentage of data points whose accuracy falls below specific precision thresholds. We assessed the methods across three different levels: errors below 1 m, 0.8 m, and 0.5 m. This percentage indicates the proportion of scenarios within the tested data that fall below the specified error thresholds.

Comparative SOTA methods: In this paper, we compare the proposed method with a CNN-based localization approach that utilizes the same extracted features for information sharing (CoLoca-CNN) [12]. Furthermore, we also compare it with a traditional point cloud matching method (PCM) that employs the Normal Distributions Transform (NDT) technique [37], where the shared information consists of the original point clouds from the vehicles. To reduce the transmission file size, feature points can be extracted and shared to facilitate point cloud matching (Feature-PCM) [32], which we include as another comparative method.

D. Localization Results

Given that it is known whether the data is transmitted from the vehicle or the infrastructure, we can utilize dedicated

TABLE II
LOCALIZATION RESULTS UNDER DIFFERENT NOISE LEVELS ON **V2Xset** (TOP) AND **V2X-Real** (BOTTOM). EACH BLOCK REPORTS V2V AND V2I WITH TWO NOISE SETTINGS ($\sigma = 2$ M / $\sigma = 1$ M)

(A) **V2Xset***V2V Localization Results with different noise levels*

Noise Level	$\sigma = 2$ m				$\sigma = 1$ m			
Methods	PCM	Feature-PCM	CoLoca-CNN	CoLoca-QuA (Ours)	PCM	Feature-PCM	CoLoca-CNN	CoLoca-QuA (Ours)
MAE (m)	2.84	2.94	0.41	0.23	0.17	0.28	0.32	0.18
RMSE (m)	2.99	3.11	0.49	0.29	0.38	0.56	0.37	0.22
error < 1 m	2.15%	2.23%	95.40%	99.43%	94.14%	99.93%	99.27%	99.73%
error < 0.8 m	1.10%	1.47%	91.92%	98.15%	93.37%	77.71%	97.74%	99.48%
error < 0.5 m	0.11%	0.45%	74.00%	93.61%	91.93%	72.78%	87.25%	97.77%

V2I Localization Results with different noise levels

Noise Level	$\sigma = 2$ m				$\sigma = 1$ m			
Methods	PCM	Feature-PCM	CoLoca-CNN	CoLoca-QuA (Ours)	PCM	Feature-PCM	CoLoca-CNN	CoLoca-QuA (Ours)
MAE (m)	2.47	2.42	0.77	0.54	0.24	0.41	0.58	0.37
RMSE (m)	2.51	2.48	0.95	0.76	0.48	0.60	0.71	0.48
error < 1 m	0.13%	0.76%	74.68%	89.89%	91.01%	87.19%	87.06%	96.67%
error < 0.8 m	0%	0.19%	61.74%	82.84%	90.18%	85.28%	76.79%	93.39%
error < 0.5 m	0%	0%	37.70%	61.80%	88.27%	81.01%	51.47%	79.18%

Voxel size: 0.2×0.2 m. X% denotes the proportion of samples below the threshold.

(B) **V2X-Real***V2V Localization Results with different noise levels*

Noise Level	$\sigma = 2$ m				$\sigma = 1$ m			
Methods	PCM	Feature-PCM	CoLoca-CNN	CoLoca-QuA (Ours)	PCM	Feature-PCM	CoLoca-CNN	CoLoca-QuA (Ours)
MAE (m)	2.03	2.49	0.99	0.61	0.71	0.67	0.88	0.57
RMSE (m)	2.38	2.67	1.18	0.78	0.99	1.17	1.01	0.72
error < 1 m	23.37%	7.52%	56.59%	85.05%	65.12%	70.71%	63.90%	86.68%
error < 0.8 m	21.77%	6.42%	44.58%	75.85%	60.29%	67.80%	50.33%	78.13%
error < 0.5 m	15.65%	5.11%	23.50%	47.98%	52.78%	65.10%	26.37%	54.44%

V2I Localization Results with different noise levels

Noise Level	$\sigma = 2$ m				$\sigma = 1$ m			
Methods	PCM	Feature-PCM	CoLoca-CNN	CoLoca-QuA (Ours)	PCM	Feature-PCM	CoLoca-CNN	CoLoca-QuA (Ours)
MAE (m)	2.76	2.61	0.30	0.19	1.27	1.28	0.30	0.16
RMSE (m)	3.33	3.02	0.46	0.23	1.85	1.61	0.45	0.19
error < 1 m	14.35%	7.85%	94.84%	99.60%	51.36%	35.43%	95.28%	96.96%
error < 0.8 m	13.45%	7.62%	94.08%	99.24%	47.08%	28.70%	94.35%	99.69%
error < 0.5 m	11.66%	6.05%	89.59%	97.69%	40.47%	20.40%	89.90%	99.11%

Voxel size: 0.4×0.4 m. X% denotes the proportion of samples below the threshold.

localization models for V2V (Vehicle-to-Vehicle) and V2I (Vehicle-to-Infrastructure) applications, respectively. Consequently, the test dataset is divided into V2V and V2I sub-datasets, and the performance is evaluated on each of these sub-datasets individually.

Results for V2V application: The quantitative localization results for V2V scenarios are summarized in Table II (Part A for V2Xset and Part B for V2X-Real). On the simulated **V2Xset**, the advantage of the proposed method is evident, particularly under large initial errors. With a noise level of $\sigma=2$ m, traditional point cloud matching methods collapse: PCM and Feature-PCM yield MAEs of 2.84 m and 2.94 m respectively, with only $\approx 2\%$ of samples falling within 1 m error. In contrast, CoLoca-QuA achieves an MAE of 0.23 m and ensures 99.43% of samples have an error < 1 m, significantly outperforming the CoLoca-CNN baseline (MAE 0.41 m). Under the smaller noise setting ($\sigma=1$ m), while PCM recovers to an MAE of 0.17 m, it requires

transmitting raw point clouds (high bandwidth). CoLoca-QuA achieves comparable accuracy (MAE 0.18 m) using only compact intermediate features.

On the real-world **V2X-Real** dataset, the trend persists despite the increased environmental complexity. Under $\sigma=2$ m noise, CoLoca-QuA achieves an MAE of 0.61 m and an RMSE of 0.78 m, reducing the error by approximately 38% compared to CoLoca-CNN (MAE 0.99 m). Furthermore, 85.05% of the inferences fall within the 1 m error threshold, whereas CoLoca-CNN only reaches 56.59%. It is observed that the performance on V2X-Real V2V is generally worse than that of V2I. This discrepancy is primarily attributed to data imbalance; in real-world data collection, V2I data is significantly more abundant than V2V data, leading to limited generalization for vehicle-to-vehicle geometric matching. Nevertheless, CoLoca-QuA still demonstrates a clear margin over the CNN-based and optimization-based baselines.

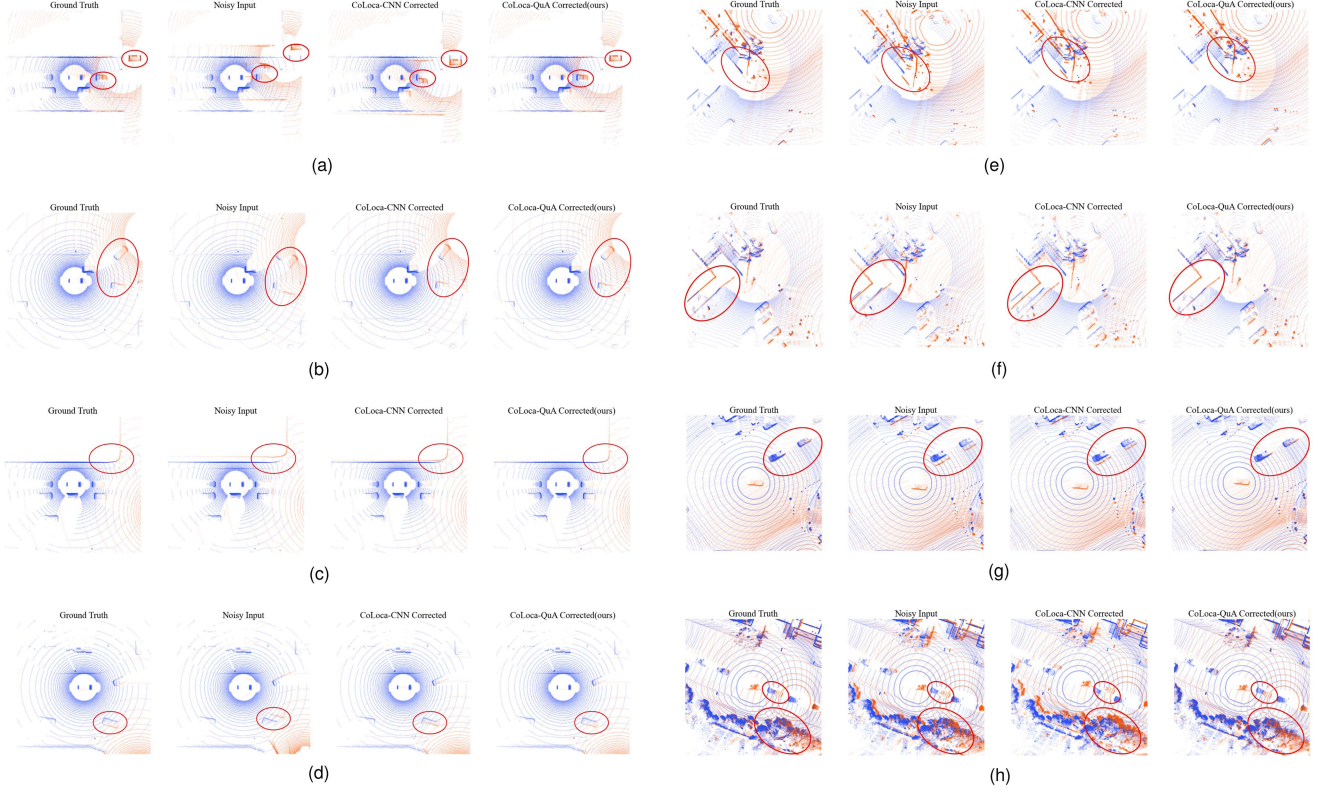


Fig. 5. Visualization results on both the simulated and real-world datasets. The left column (subfigures (a)–(d)) shows four representative scenarios from the V2XSet simulated dataset, while the right column (subfigures (e)–(h)) shows the scenarios from the V2X-Real real-world dataset. In all cases, the ego vehicle’s point cloud is in blue and that from the other agent in orange. Each subfigure compares four settings: the ideal Ground Truth alignment, the Noisy Input with a significant initial pose error, the result after correction by CoLoca-CNN, and the result from our proposed CoLoca-QuA (ours). The red circles highlight critical regions where objects are severely misaligned in the noisy input but are accurately re-aligned by our model. (a) V2XSet: V2I scenario 1. (b) V2XSet: V2I scenario 2. (c) V2XSet: V2V scenario 1. (d) V2XSet: V2V scenario 2. (e) V2X-Real: V2I scenario 1. (f) V2X-Real: V2I scenario 2. (g) V2X-Real: V2V scenario 1. (h) V2X-Real: V2V scenario 2.

Results for V2I application: The results for V2I application, presented in Table II, further validate the robustness of the proposed approach. On **V2Xset** with $\sigma=2$ m, PCM-based methods fail completely (0% or nearly 0% success rate at 0.5 m). CoLoca-QuA, however, maintains high precision with an MAE of 0.54 m, surpassing CoLoca-CNN (0.77 m).

Most notably, on the **V2X-Real** dataset, CoLoca-QuA exhibits exceptional performance. Under $\sigma=2$ m noise, it achieves an MAE of 0.19 m and an RMSE of 0.23 m. In comparison, CoLoca-CNN yields an MAE of 0.30 m, and PCM degrades to 2.76 m. The robustness of our method is highlighted by the threshold metrics: 99.60% of the test samples have an error less than 1 m, and 97.69% are within 0.5 m. Even under the strict bandwidth constraint (using 0.4×0.4 m voxels for real-world data), CoLoca-QuA effectively captures global geometric relationships that local convolution (CNN) and iterative registration (PCM) struggle to resolve under large displacements. Overall, across both datasets and noise levels, the query-based head achieves the best accuracy-efficiency trade-off, particularly in regimes with large initial localization errors where traditional pipelines struggle to converge.

Visualization: Across both V2Xset and V2X-Real, the proposed CoLoca-QuA maintains strong localization performance even under large initial pose errors and consistently outperforms

CoLoca-CNN on all reported metrics. This advantage holds in both V2V and V2I configurations, indicating that the query-based transformer head is effective at recovering substantial misalignments before downstream fusion. Visualizations shown in Fig. 5 further illustrate this behavior: although the transformed CAV point cloud (red) initially misaligns with the ego point cloud (blue) due to large initial pose errors, our method accurately estimates and corrects the relative pose so that the point clouds from two agents align well after refinement.

E. Complexity and Latency Analysis

Complexity Analysis: We denote the input feature map dimensions as $C_{in} \times H \times W$. The computational costs for the two proposed heads are formulated as follows:

a) **CoLoca-QuA (Transformer):** Given a patch size P (yielding $N = HW/P^2$ tokens), embedding dimension D , network depth L , and MLP expansion ratio r , the total floating-point operations (FLOPs) scale as:

$$\underbrace{\mathcal{O}(HWC_{in}D)}_{\text{Patch Embedding}} + \underbrace{\mathcal{O}(L(N^2D + rND^2))}_{\text{Transformer Encoder}} \quad (15)$$

b) **CoLoca-CNN:** Assuming a constant channel width C maintained across K blocks, where each block contains a 3×3

TABLE III
COMPARISON OF COMPLEXITY AND LATENCY

Method	Complexity (GMACs)	Params (M)	Latency (ms)
CoLoca-QuA	3.70	48.74	41.07
CoLoca-CNN	77.24	12.85	40.00

convolution and a pooling layer, the cost is dominated by the convolutions:

$$\sum_{k=1}^K \mathcal{O}(H_k W_k C^2), \quad (16)$$

where H_k, W_k represent the spatial resolution at the k -th stage. The subsequent fully connected layers contribute negligible lower-order terms.

To evaluate the proposed methods, we utilize the following concrete hyperparameters:

- *Input*: $512 \times 96 \times 256$ ($C_{in} = 512$).
- *CoLoca-QuA*: $P = 16$ (resulting in $N = 96$ tokens), $D = 256$, $L = 6$, $r = 4$.
- *CoLoca-CNN*: 5 blocks with $C = 512$ and 2×2 average pooling per block.

The measured performance metrics—computational cost (GMACs), model size (Parameters), and inference latency—are summarized in Table III. Note that MAC counts refer strictly to the pose-regression heads, excluding the shared backbone.

As shown in Table III, CoLoca-CNN exhibits a substantially higher theoretical computational cost (77.24 GMACs) compared to CoLoca-QuA (3.70 GMACs). However, the wall-clock latency is nearly identical (≈ 40 ms). This discrepancy is attributed to the high parallelization efficiency of convolution kernels on modern GPUs and the aggressive spatial downsampling in the CNN head, which rapidly reduces activation memory traffic compared to the attention mechanisms in the Transformer. Given a typical LiDAR sensor rate of 10 Hz (a 100 ms period), both heads satisfy real-time constraints with ample margin. Therefore, the latency of CoLoca-QuA is acceptable in real-world practice.

F. Ablation Study

Weighting parameters of the loss function: The training objective is governed by the loss function formulated in (14). In this section, we analyze the sensitivity of the model performance to the hyperparameters $(\alpha_x, \alpha_y, \alpha_\varphi)$ defined in (14). We ultimately adopt the weighting combination $(\alpha_x, \alpha_y, \alpha_\varphi) = (2, 2, 1)$ for the weighted MSE loss. This selection follows the principle of balancing the magnitudes of translational and rotational residuals, a strategy consistent with prior practice in [34]. To validate this configuration, we conducted an ablation study comparing three different parameter settings:

- S1**: $(\alpha_x, \alpha_y, \alpha_\varphi) = (1, 1, 2)$,
- S2**: $(\alpha_x, \alpha_y, \alpha_\varphi) = (1, 1, 1)$,
- S3**: $(\alpha_x, \alpha_y, \alpha_\varphi) = (2, 2, 1)$.

The comparative results are presented in Fig. 6. As illustrated, **S3** yields the lowest RMSE for both V2I and V2V scenarios. Furthermore, Fig. 6(b) confirms that this gain in translational accuracy does not compromise heading estimation, as the mean

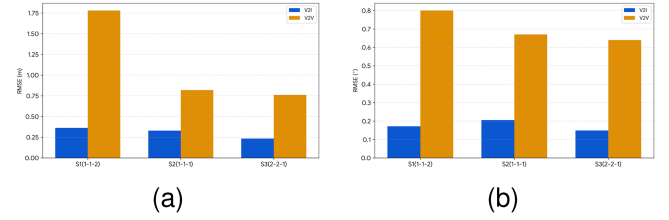


Fig. 6. RMSE of pose error under different weighting parameters of the loss function. (a) Translation error. (b) Yaw error.

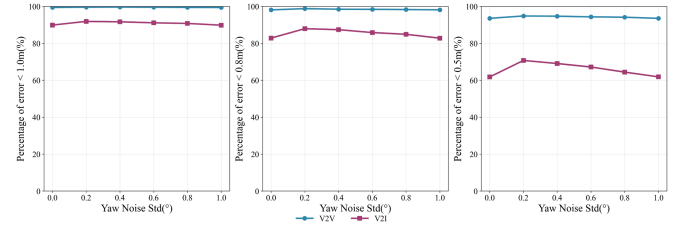


Fig. 7. Sensitivity analysis of the model with respect to the initial yaw error. The accuracy is evaluated on both V2V and V2I datasets under three different error thresholds.

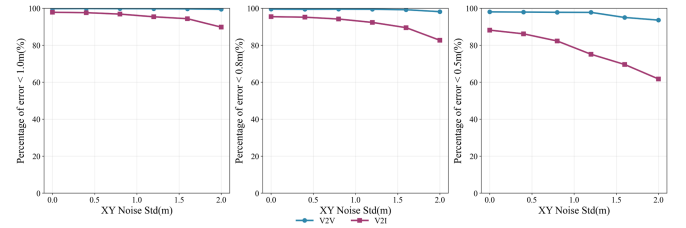


Fig. 8. Sensitivity analysis of the model with respect to the initial translational (XY) error. A noticeable performance drop is observed, especially for the V2I case.

yaw errors under **S3** remain highly competitive. Consequently, the parameter setting $(\alpha_x, \alpha_y, \alpha_\varphi) = (2, 2, 1)$ is utilized for the final loss function formulation.

Robustness assessment on initial pose errors: To rigorously evaluate the robustness of our proposed CoLoca-QuA model, we conducted a sensitivity analysis by systematically introducing initial pose errors. Since system performance is critically dependent on handling such inaccuracies, we designed two distinct experimental protocols to isolate the effects of translational and rotational (yaw) errors.

To assess sensitivity to translational error, we varied its standard deviation over $[0, 2]$ m while maintaining a constant yaw error 2° . Conversely, to evaluate the impact of rotational error, we varied the yaw's standard deviation within $[0, 1]^\circ$ while fixing the translational error at 2 m. All evaluations were performed on the V2V and V2I datasets. The results of this sensitivity analysis are presented in Figs. 7 and 8.

Fig. 7 illustrates the model's performance under varying initial yaw errors. It is evident that the model is highly robust to this type of noise. For the V2V scenario (blue curve), performance remains almost perfectly stable across all tested yaw noise levels, maintaining accuracy above 95% even for the strict error threshold of 0.5 m. Similarly, the V2I performance (purple curve) shows only a marginal decline as the yaw noise

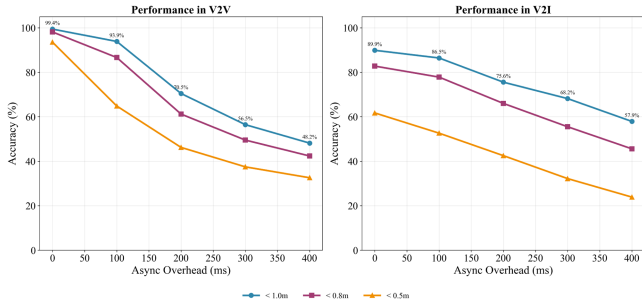


Fig. 9. Sensitivity assessment on time delay.

TABLE IV
TRANSMISSION FILE SIZE

Methods	PCM	Feature-PCM	CoLoca-CNN	CoLoca-QuA(Ours)
File size (Mb)	2	0.2	0.1	0.1

standard deviation increases from 0° to 1.0° . This demonstrates the model's strong invariance to initial rotational inaccuracies.

In contrast, Fig. 8 reveals a relatively greater sensitivity to initial translational errors. A clear downward trend in accuracy is observed for both V2V and V2I scenarios as the standard deviation of the translational noise increases. This performance degradation is particularly pronounced in the V2I case. For instance, at the 0.5 m error threshold, the V2I accuracy drops significantly from approximately 88% to 61% as the noise standard deviation increases from 0 to 2 m. While the V2V performance also decreases, its degradation is more graceful.

Collectively, these findings indicate that our CoLoca-QuA model is substantially more sensitive to initial translational errors than to initial rotational (yaw) errors, especially for the V2I scenarios.

Sensitivity analysis on time delay: We also evaluated the impact of asynchronous time delays over a range of [0, 400] ms, which is a critical factor in real-world deployments. As shown in Fig. 9, the model's accuracy is highly sensitive to communication latency.

The primary finding is a consistent and sharp performance degradation across all evaluated thresholds in both V2V and V2I scenarios as the delay increases. This is due to the growing spatio-temporal misalignment between agents' data caused by latency. In the V2V case, for example, the accuracy at the 0.5 m threshold falls drastically from 93.4% to 32.5% when the delay rises to 400 ms. Although the V2I scenario starts with lower baseline accuracy, it exhibits a similar downward trend. Interestingly, for less stringent accuracy requirements (e.g., < 1.0 m), V2I performance degrades more gradually than V2V at very high delays, eventually surpassing it at 400 ms. This indicates that the stability of infrastructure sensors may offer some resilience to high latency where coarse localization is sufficient.

Given this significant sensitivity to time delay, efficient data transmission is a key consideration in V2X applications. The intermediate feature-sharing mechanism requires the lowest bandwidth, as demonstrated in Table IV. The transmission file size for the learning-based methods, including the proposed CoLoca-QuA and CoLoca-CNN, is approximately 0.1 MB. In contrast, the file size for Feature-PCM is around 0.2 MB, while PCM requires about 2 MB. Thus, the proposed method not only

achieves better accuracy but also demands the lowest bandwidth, making it ideal for multi-agent cooperative driving tasks.

V. CONCLUSIONS AND FUTURE WORK

This work presents the first cooperative relative pose estimation framework that utilizes shared intermediate neural features among multiple agents within a Transformer-based architecture, introducing a novel query-based learning network. The proposed query-based method effectively balances accuracy and efficiency by employing attention mechanisms across the entire feature map of the sensing area, which significantly enhances relative pose estimation accuracy. Unlike traditional point cloud matching methods, our approach robustly handles scenarios with large initial localization errors, thereby offering substantial benefits for cooperative driving automation.

Future work will focus on integrating our network into real-world, real-time physical systems to improve the performance of cooperative perception. In a typical intermediate-fusion pipeline, the BEV features already computed for cooperative perception can be directly used as inputs for CoLoca-QuA to regress the relative pose error between agents. The estimated pose can then be used to realign the CAV feature map via a differentiable BEV warping operator. The aligned CAV feature map and the ego feature map can subsequently be fed into downstream detection heads or other networks without modifying their interfaces. Consequently, the relative pose estimate not only is derived from intermediate features but also feeds back to correct their spatial alignment prior to cooperative detection. Therefore, our method, based on sharing intermediate neural features, shows significant potential not only for achieving high relative pose estimation accuracy but also for seamless integration into existing intermediate feature fusion-based cooperative perception frameworks. Additionally, we will further investigate a more comprehensive framework capable of handling multiple agents' information to output all relative poses between each agent and the ego vehicle.

REFERENCES

- [1] R. Xu et al., "The OpenCDA open-source ecosystem for cooperative driving automation research," *IEEE Trans. Intell. Veh.*, vol. 8, no. 4, pp. 2698–2711, Apr. 2023.
- [2] W. Li, M. Liu, T. Chen, and G. Mao, "Vehicles selection algorithm for cooperative localization based on stochastic geometry in internet of vehicle systems," *IEEE Trans. Veh. Technol.*, vol. 74, no. 3, pp. 4893–4903, Mar. 2025.
- [3] H. Zhang et al., "Evaluation of eco-driving performance of electric vehicles using driving behavior-enabled graph spectrums: A naturalistic driving study in China," *Green Energy Intell. Transp.*, vol. 4, no. 1, 2025, Art. no. 100246. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2773153724000987>
- [4] S. Kitajima, H. Chouchane, J. Antona-Makoshi, N. Uchida, and J. Tajima, "A nationwide impact assessment of automated driving systems on traffic safety using multiagent traffic simulations," *IEEE Open J. Intell. Transp. Syst.*, vol. 3, pp. 302–312, 2022.
- [5] D. Chen, Y. Kim, and A. G. Stefanopoulou, "State of charge node planning with segmented traffic information," in *Proc. 2018 Annu. Amer. Control Conf.*, 2018, pp. 4969–4974.
- [6] M. Nandhini and M. M. Rabik, "A comprehensive overview of the alignment between platoon control approaches and clustering strategies," *Green Energy Intell. Transp.*, vol. 3, no. 6, 2024, Art. no. 100223. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2773153724000756>

- [7] T. Li, J. Ruan, and K. Zhang, "The investigation of reinforcement learning-based end-to-end decision-making algorithms for autonomous driving on the road with consecutive sharp turns," *Green Energy Intell. Transp.*, vol. 4, no. 3, 2025, Art. no. 100288, [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2773153725000386>
- [8] B. Lu, X. Huang, Y. Wu, L. Qian, S. Zhou, and D. Niyato, "Joint optimization of compression, transmission and computation for cooperative perception aided intelligent vehicular networks," *IEEE Trans. Veh. Technol.*, vol. 74, no. 5, pp. 8201–8214, May 2025.
- [9] R. Xu, H. Xiang, X. Xia, X. Han, J. Li, and J. Ma, "OPV2V: An open benchmark dataset and fusion pipeline for perception with vehicle-to-vehicle communication," in *Proc. 2022 Int. Conf. Robot. Automat.*, 2022, pp. 2583–2589.
- [10] Y. Lu et al., "Robust collaborative 3 D object detection in presence of pose errors," in *Proc. 2023 IEEE Int. Conf. Robot. Automat.*, 2023, pp. 4812–4818.
- [11] Z. Song, F. Wen, H. Zhang, and J. Li, "A cooperative perception system robust to localization errors," in *Proc. 2023 IEEE Intell. Veh. Symp. (IV)*, 2023, pp. 1–6.
- [12] L. Gao, H. Xiang, X. Xia, and J. Ma, "End-to-end cooperative localization via neural feature sharing," in *Proc. 2024 IEEE Intell. Veh. Symp. (IV)*, 2024, pp. 553–558.
- [13] X. Li et al., "A cooperative relative localization system for distributed multi-agent networks," *IEEE Trans. Veh. Technol.*, vol. 72, no. 11, pp. 14828–14843, Nov. 2023.
- [14] H. Chi, P. Cai, D. Fu, J. Zhai, Y. Zeng, and B. Shi, "Spatiotemporal-restricted algorithm as a support for lane-free traffic at intersections with mixed flows," *Green Energy Intell. Transp.*, vol. 3, no. 2, 2024, Art. no. 100159, [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2773153724000112>
- [15] J. Zhang, R. Zhan, Y. Wang, and X. Qu, "Optical communication based V2V for vehicle platooning," *Green Energy Intell. Transp.*, vol. 4, no. 4, 2025, Art. no. 100278, [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2773153725000283>
- [16] D. Kumar and N. Muhammad, "A survey on localization for autonomous vehicles," *IEEE Access*, vol. 11, pp. 115865–115883, 2023.
- [17] T. Ando, W. Kugimiya, T. Hashimoto, F. Momiyama, K. Aoki, and K. Nakano, "Lateral control in precision docking using RTK-GNSS/INS and LiDAR for localization," *IEEE Trans. Intell. Veh.*, vol. 6, no. 1, pp. 78–87, Mar. 2021.
- [18] A. E. Mahdi, A. Azouz, A. Abdalla, and A. Abosekeen, "IMU-error estimation and cancellation using ANFIS for improved UAV navigation," in *Proc. 2022 13th Int. Conf. Elect. Eng.*, 2022, pp. 120–124.
- [19] M. A. Ghadiri-Modarres, M. Mojiri, and H. R. Z. Zangeneh, "Nonholonomic source localization in 3-D environments without position measurement," *IEEE Trans. Autom. Control*, vol. 61, no. 11, pp. 3563–3567, Nov. 2016.
- [20] G. Elghazaly, R. Frank, S. Harvey, and S. Safko, "High-definition maps: Comprehensive survey, challenges, and future perspectives," *IEEE Open J. Intell. Transp. Syst.*, vol. 4, pp. 527–550, 2023.
- [21] L. Gao, L. Xiong, X. Xia, Y. Lu, Z. Yu, and A. Khajepour, "Improved vehicle localization using on-board sensors and vehicle lateral velocity," *IEEE Sensors J.*, vol. 22, no. 7, pp. 6818–6831, Apr. 2022.
- [22] M. Sakthi, M. Arvinte, and H. Vikalo, "Automotive radar sub-sampling via object detection networks: Leveraging prior signal information," *IEEE Open J. Intell. Transp. Syst.*, vol. 4, pp. 858–869, 2023.
- [23] Y. Zhang, Z. Fan, J. Hou, N. Chen, F. Lyu, and P. Yang, "Incentivizing point cloud-based accurate cooperative perception for connected vehicles," *IEEE Trans. Veh. Technol.*, vol. 74, no. 4, pp. 5637–5648, Apr. 2025.
- [24] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, "PointNet: Deep learning on point sets for 3 D classification and segmentation," in *Proc. IEEE Conf. Comput. Vis., Pattern Recognit.*, 2017, pp. 652–660.
- [25] Y. Zhou and O. Tuzel, "VoxelNet: End-to-end learning for point cloud based 3 D object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 4490–4499.
- [26] Y. Yan, Y. Mao, and B. Li, "SECOND: Sparsely embedded convolutional detection," *Sensors*, vol. 18, no. 10, 2018, Art. no. 3337.
- [27] N. Deo and M. M. Trivedi, "Convolutional social pooling for vehicle trajectory prediction," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, 2018, pp. 1468–1476.
- [28] A. H. Lang, S. Vora, H. Caesar, L. Zhou, J. Yang, and O. Beijbom, "PointPillars: Fast encoders for object detection from point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 12697–12705.
- [29] R. Xu, H. Xiang, Z. Tu, X. Xia, M.-H. Yang, and J. Ma, "V2X-ViT: Vehicle-to-everything cooperative perception with vision transformer," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 107–124.
- [30] J. Xin, G. Xie, B. Yan, M. Shan, P. Li, and K. Gao, "Multimobile robot cooperative localization using ultrawideband sensor and GPU acceleration," *IEEE Trans. Automat. Sci. Eng.*, vol. 19, no. 4, pp. 2699–2710, Oct. 2022.
- [31] J. Dong et al., "LiDAR-based cooperative relative localization," in *Proc. 2023 IEEE Intell. Veh. Symp. (IV)*, 2023, pp. 1–8.
- [32] L. Gao, H. Xiang, X. Xia, and J. Ma, "Multisensor fusion for vehicle-to-vehicle cooperative localization with object detection and point cloud matching," *IEEE Sensors J.*, vol. 24, no. 7, pp. 10865–10877, Apr. 2024.
- [33] J. Gu et al., "FeaCo: Reaching robust feature-level consensus in noisy pose conditions," in *Proc. 31st ACM Int. Conf. Multimedia*, 2023, pp. 3628–3636.
- [34] N. Vadivelu, M. Ren, J. Tu, J. Wang, and R. Urtasun, "Learning to communicate and correct pose errors," in *Proc. Conf. Robot Learn.*, 2021, pp. 1195–1210.
- [35] H. Xiang et al., "V2X-real: A large-scale dataset for vehicle-to-everything cooperative perception," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 455–470, doi: [10.1007/978-3-031-72943-0_26](https://doi.org/10.1007/978-3-031-72943-0_26).
- [36] Q. Chen, X. Ma, S. Tang, J. Guo, Q. Yang, and S. Fu, "F-cooper: Feature based cooperative perception for autonomous vehicle edge computing system using 3 D point clouds," in *Proc. 4th ACM/IEEE Symp. Edge Comput.*, 2019, pp. 88–100.
- [37] P. Biber and W. Straßer, "The normal distributions transform: A new approach to laser scan matching," in *Proc. 2003 IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS 2003)(Cat. No 03CH37453)*, 2003, vol. 3, pp. 2743–2748.



Letian Gao (Member, IEEE) received the B.E. and Ph.D. degrees in vehicle engineering from the School of Automotive Studies, Tongji University, Shanghai, China, in 2016 and 2021, respectively. He was a Visiting Student with the Department of Mechanical and Mechatronics Engineering, University of Waterloo, Waterloo, ON, Canada, an Algorithm Engineer with SAIC Volkswagen, Shanghai, and a Postdoctoral Researcher with the Department of Civil and Environmental Engineering, University of California, Los Angeles, Los Angeles, CA, USA. He is currently an Assistant Professor with the School of Automotive Studies, Tongji University. His research interests include cooperative perception, vehicle localization, state estimation, and dynamics control of intelligent vehicles.



Hao Xiang received the B.S. degree in microelectronics from Fudan University, Shanghai, China, in 2019, the M.S. degree in machine learning and data science from the University of California, San Diego, San Diego, CA, USA, in 2021, and the Ph.D. degree with the UCLA Mobility Lab, University of California, Los Angeles, Los Angeles, CA, in 2024. His research interests include computer vision, multiagent cooperative perception, multimodal sensor fusion, trajectory prediction, robotics, and autonomous driving.



Tianhao Tan received the B.E. degree in vehicle engineering from the Wuhan University of Technology, Wuhan, China in 2025. He is currently working toward the M.S. degree with the School of Automotive Studies, Tongji University, Shanghai, China. His research interests include cooperative perception, vehicle localization, and state estimation.



Zonglin Meng received the B.S. degree in computer science from the University of Wisconsin Madison, Madison, WI, USA, in 2019, the Ph.D. degree with UCLA Mobility Lab, University of California at Los Angeles, Los Angeles, CA, USA in 2024. He is currently a Postdoctoral Researcher with the Department of Civil and Environmental Engineering, UCLA, Los Angeles. His research interests include cooperative detection, cooperative tracking, and multiagent learning.



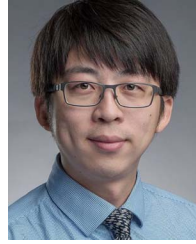
Xin Xia (Senior Member, IEEE) received the B.E. degree in vehicle engineering from the School of Mechanical and Automotive Studies, South China University of Technology, Guangzhou, China, in 2014, and the Ph.D. degree in vehicle engineering from the School of Automotive Studies, Tongji University, Shanghai, China, in 2019. He was a Research Scientist with the Department of Civil and Environmental Engineering, University of California, Los Angeles, CA, USA. He was a Postdoctoral Fellow associated with Dr. Amir Khajepour with the Department of

Mechanical and Mechatronics Engineering, University of Waterloo, Waterloo, ON, Canada, from 2020 to 2021. He is currently an Assistant Professor with Department of Mechanical Engineering, University of Michigan, Dearborn, MI, USA. His research interests include state estimation, cooperative localization, cooperative perception, and dynamics control of the autonomous vehicle.



Lu Xiong received the Ph.D. degree in vehicle engineering from Tongji University, Shanghai, China, in 2005. He is currently the Dean and a Professor with the School of Automotive Studies, Tongji University. He is also the Associate Director of the Clean Energy Automotive Engineering Center, Tongji University, which is the National Engineering Laboratory for New Energy Vehicle and Powertrain of China. His research interests include the dynamic control of distributed drive electric vehicles, motion planning and control of intelligent vehicles, and all-terrain vehicles.

He was the recipient of the Fund for Distinguished Young Scholars (National Science Foundation of China) and the First Prize in the Shanghai Science and Technology Progress Awards in 2013, 2020, and 2022.



Jiaqi Ma (Senior Member, IEEE) received the Ph.D. degree in transportation engineering from the University of Virginia, Charlottesville, VA, USA, in 2014. He is currently an Associate Professor with the Samueli School of Engineering, University of California, Los Angeles, Los Angeles, CA, USA, and the Director of the Federal Highway Administration Center of Excellence on New Mobility and Automated Vehicles. He has led and managed numerous research projects funded by U.S. Department of Transportation (DOT), National Science Foundation, state DOTs,

and other federal/state/local programs covering areas of smart transportation systems, such as vehicle-highway automation, intelligent transportation systems, connected vehicles, shared mobility, large-scale smart system modeling and simulation, and artificial intelligence and advanced computing applications in transportation. He is the Editor-in-Chief of IEEE OPEN JOURNAL OF INTELLIGENT TRANSPORTATION SYSTEMS. He is also a Member of the Transportation Research Board (TRB) Standing Committee on Vehicle-Highway Automation, the TRB Standing Committee on Artificial Intelligence and Advanced Computing Applications, and the American Society of Civil Engineers Connected and Autonomous Vehicles Impacts Committee, and he is the Cochair of the IEEE Intelligent Transportation Systems Society Technical Committee on Smart Mobility and Transportation 5.0.